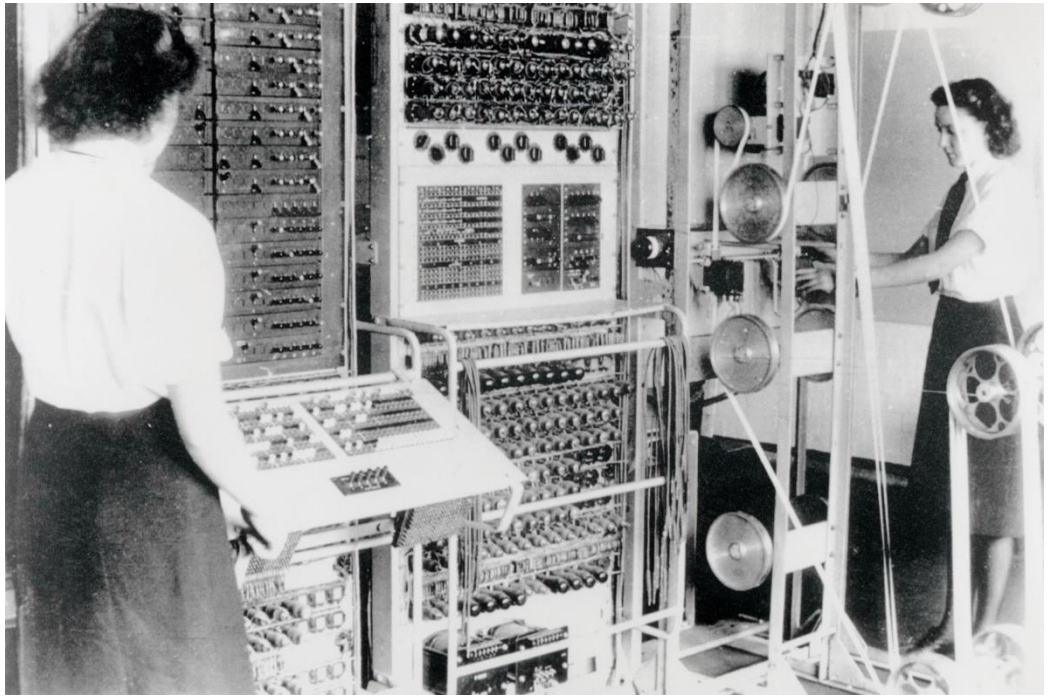
# CSCI 210: Computer Architecture
# Lecture 19: Clocks, Latches, and Flip Flops

Stephen Checkoway

Slides from Cynthia Taylor

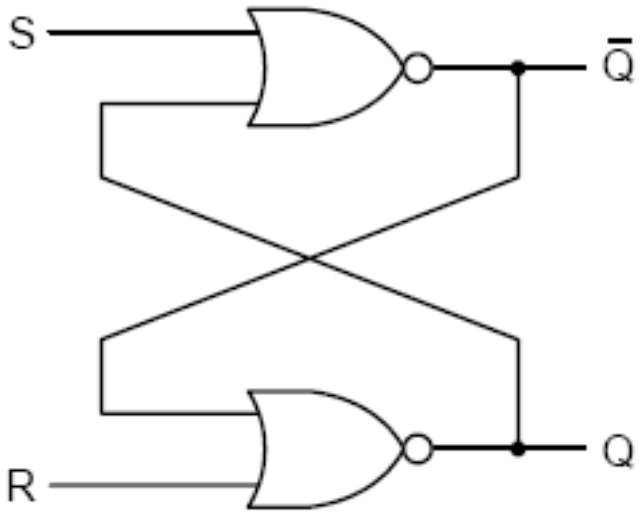
# CS History: Latches



A Colossus Mark 2 codebreaking computer being operated by Dorothy Du Boisson (left) and Elsie Booker (right), 1943

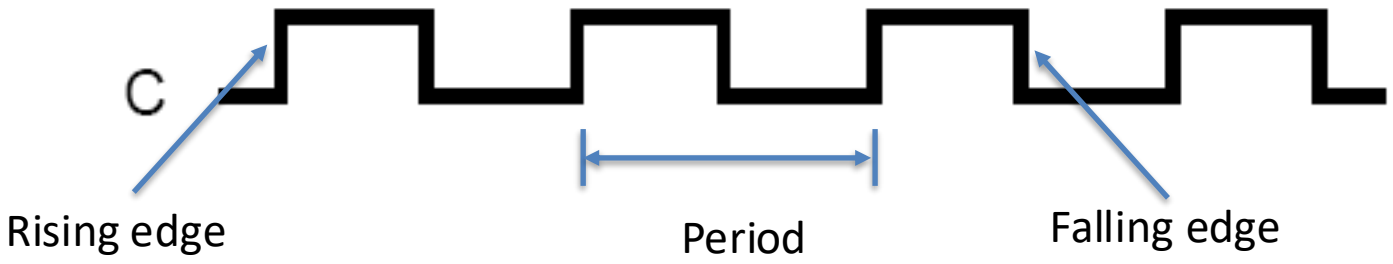
- The first electronic latch was invented in 1918 by British physicists William Eccles and F. W. Jordan.
- Used in the 1943 British Colossus codebreaking computer (made out of vacuum tubes)
- Modern flip flops made of logic gates were first discussed in a 1954 UCLA course on computer design by Montgomery Phister

# S-R Latch




- Set (S=1, R=0): $Q_{next} = 1$
- Reset (S=0, R=1): $Q_{next} = 0$
- Neither (S=0, R=0): $Q_{next} = Q_{curr}$

# Clock




- Oscillates between 1 and 0 with a fixed period
  - 0 to 1 transition is a **rising edge**
  - 1 to 0 transition is a **falling edge**
  - Time between two rising (or falling) edges is one **period** or **cycle**
- Used to control when values change

# Clocked SR Latch

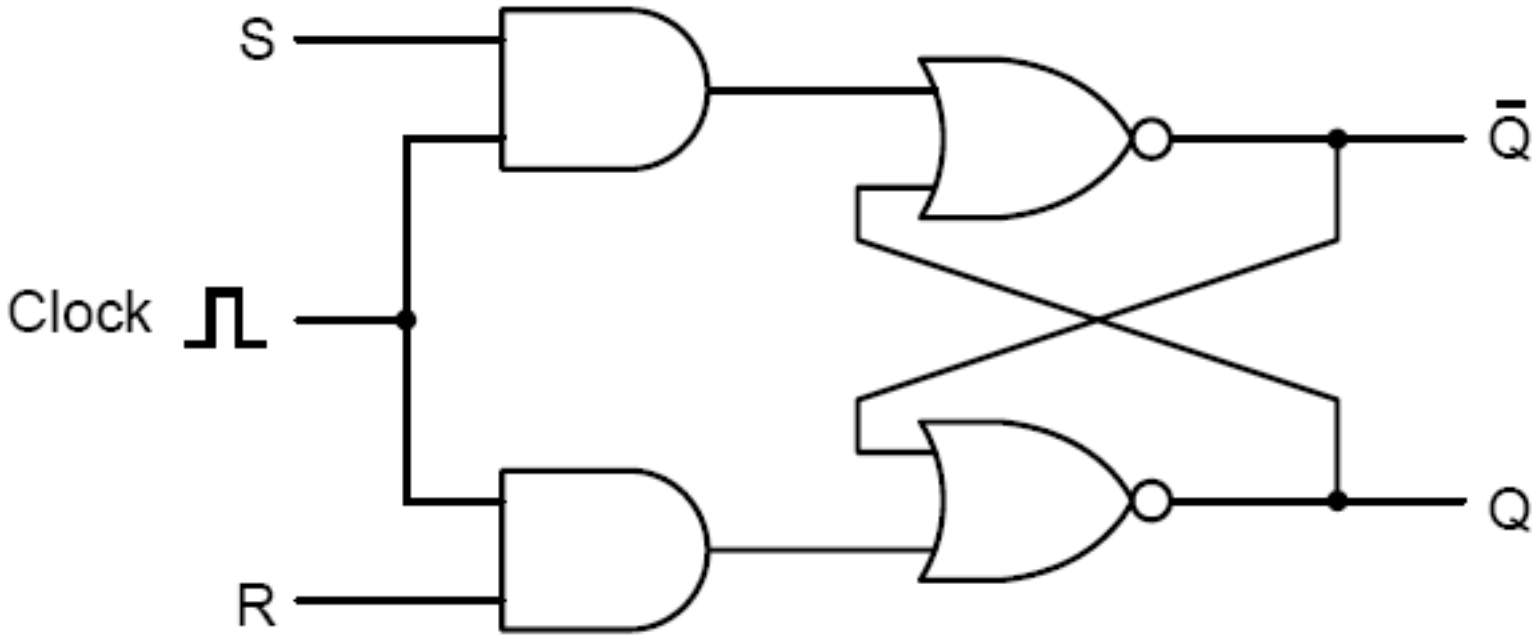



**Figure 3-23.** A clocked SR latch.

- Only changes state when the clock is asserted (meaning the clock has value 1)

# Given S, R and Clock, Q will be:

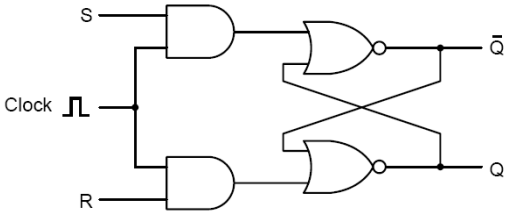
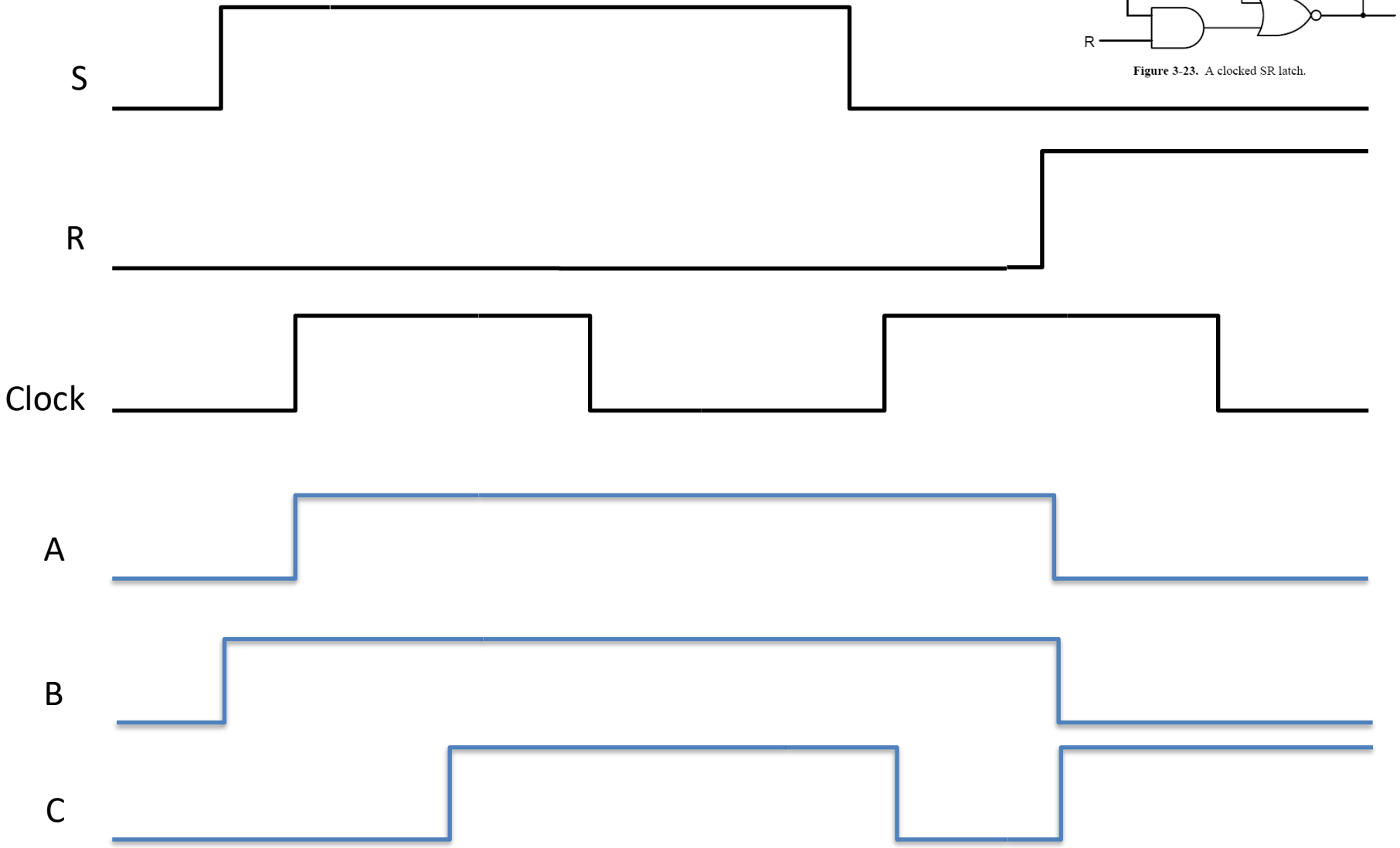
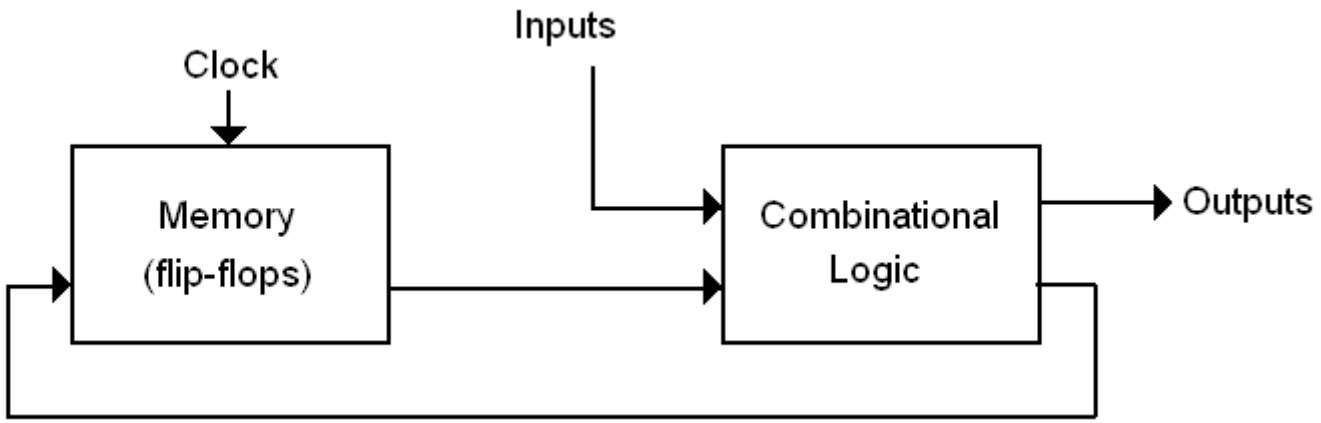



**Figure 3-23.** A clocked SR latch.



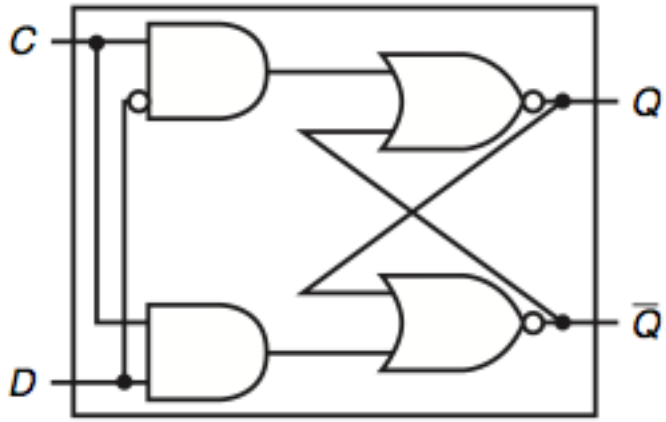

D. None of the above

# Why Clock a Latch?



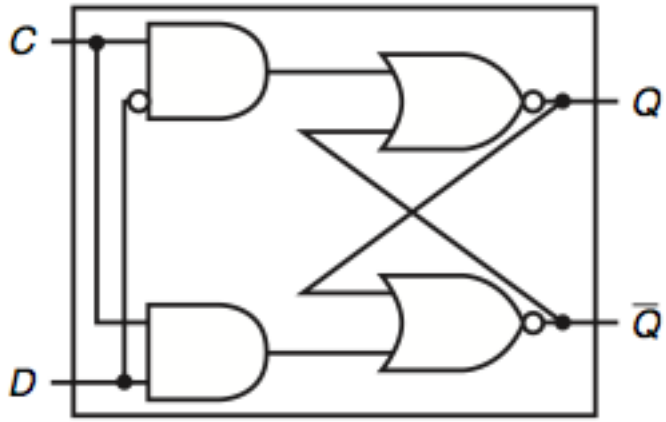

- Can save the results of combinational logic (think the ALU)
- If a latch is clocked, we know the values in it won't change as we perform combinational logic on them
  - Think of performing addi t0, t0, 1
- In today's class, we'll build memory that only changes values at one precise instant

# Clocked D-Latch




- S-R latch, but now there is a single input, D, ANDed with the clock C
- Now impossible to have both inputs set to 1

# Which Column Completes the Truth Table?

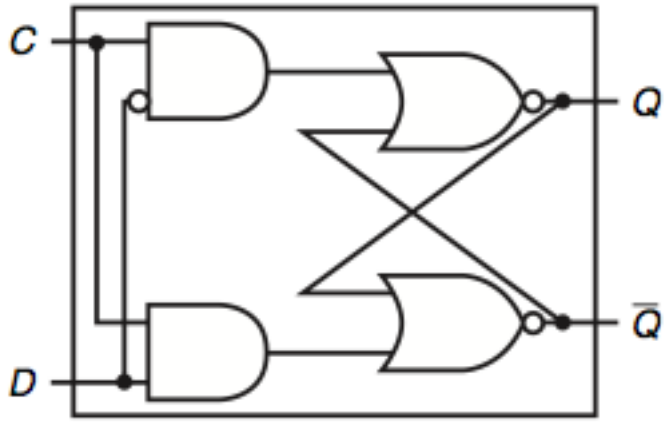



| C | D | $Q_{next}$ |
|---|---|---|
| 1 | 1 | |
| 1 | 0 | |

| A | B | C | D |
|---|---|---|---|
| 1 | 1 | 0 | 1 |
| 1 | 0 | 1 | $Q_{curr}$ |

E. None of the above

# Which Column Completes the Truth Table?

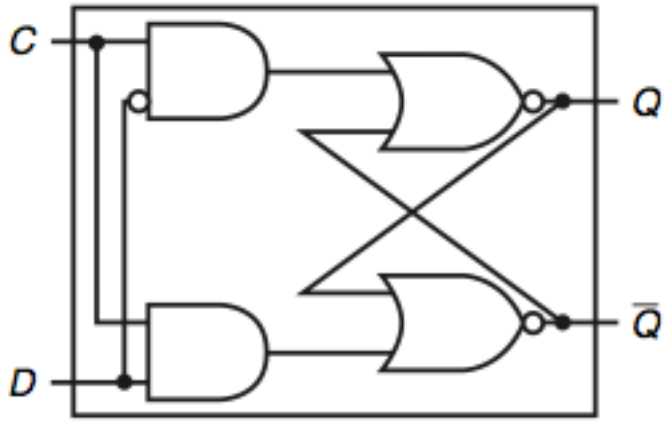



| C | D | $Q_{next}$ |
|---|---|---|
| 0 | 1 | |
| 0 | 0 | |

| A | B | C | D |
|---|---|---|---|
| 0 | 1 | 1 | $Q_{curr}$ |
| 0 | 0 | $Q_{curr}$ | $Q_{curr}$ |

E. None of the above

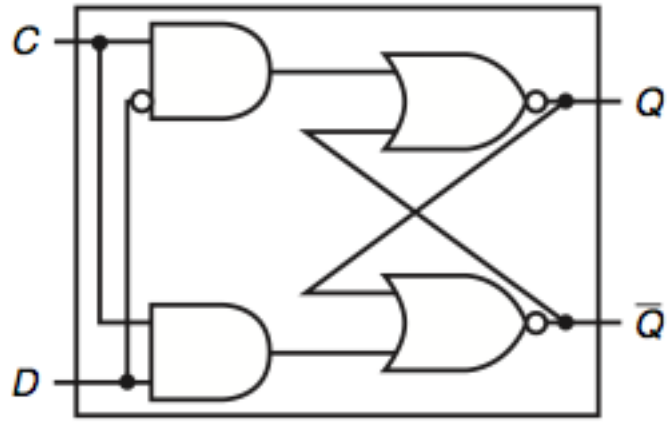
# Clocked D-Latch

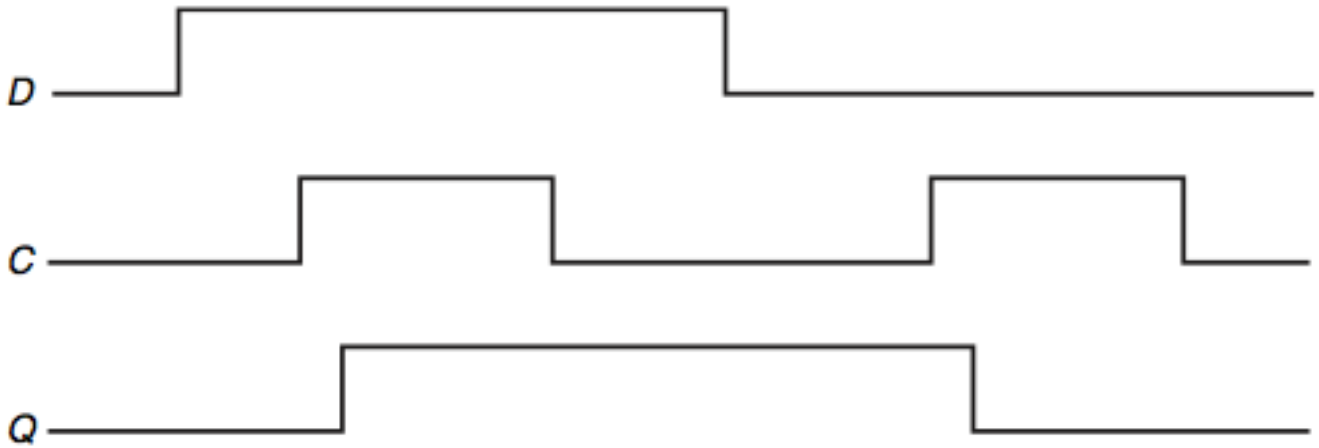



- Latch is “open” when clock is asserted (asserted = logical 1)

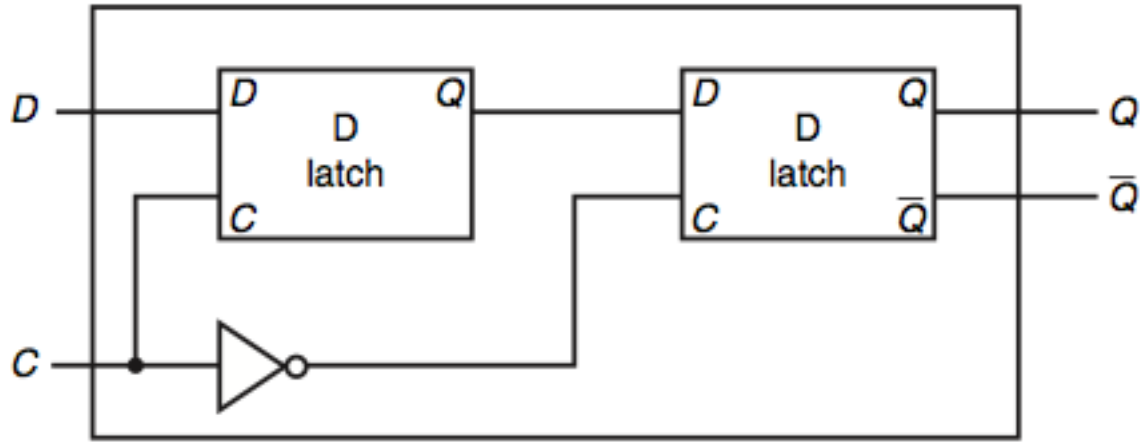
- Q = value of D when the latch is open
  Q = most recently set value when the latch is closed

# Clocked D-Latch; note output takes a little time to change after the clock goes high



# D Flip-Flop




- Two D-Latches, with the clock negated to the second latch

# The value of (the right-most) Q in the flip flop can change

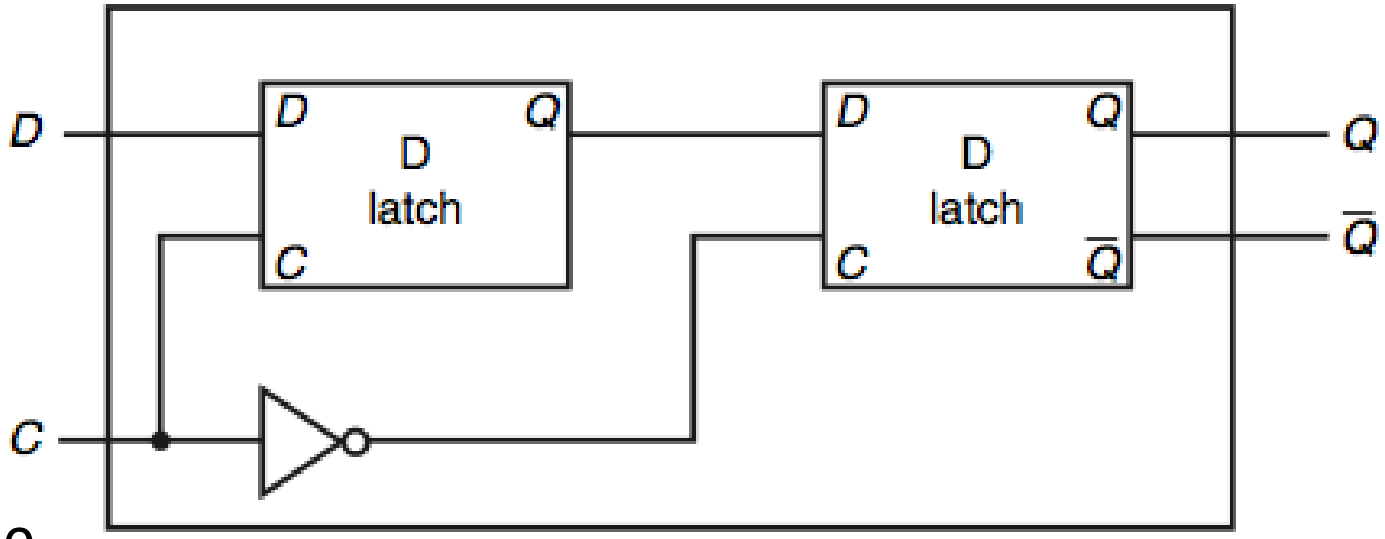
A. Any time the clock is 1.

B. Any time the clock is 0.

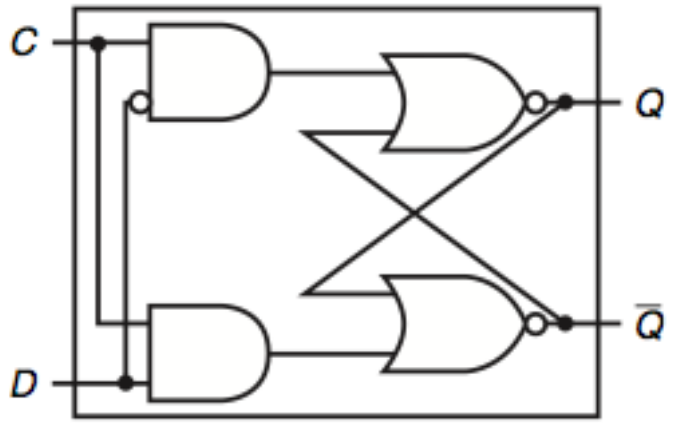
C. When the clock changes from 1 to 0.

D. When the clock changes from 0 to 1.

E. None of the above

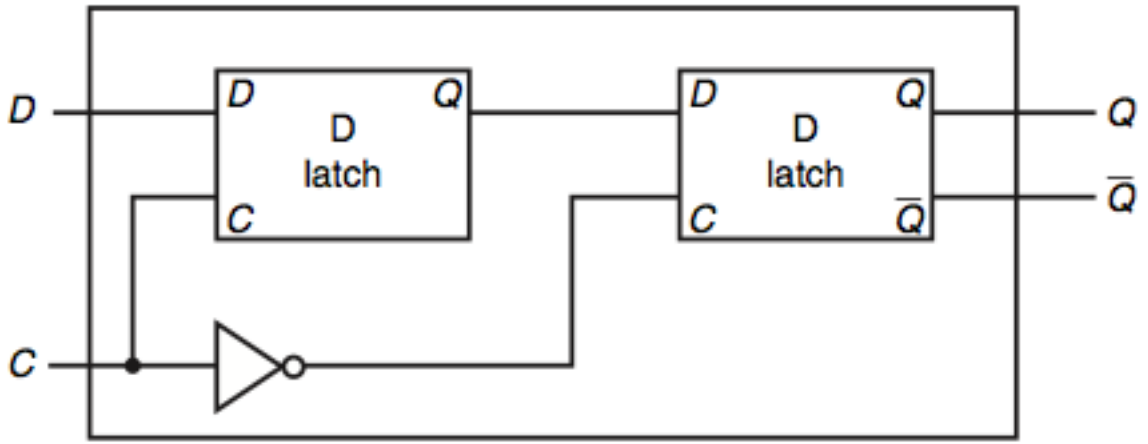


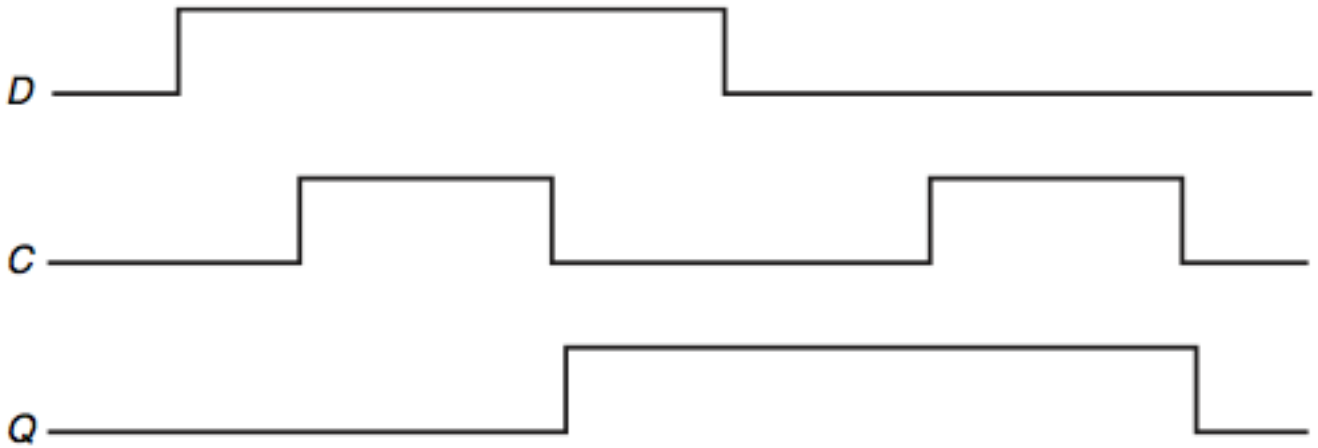

D FlipFlop



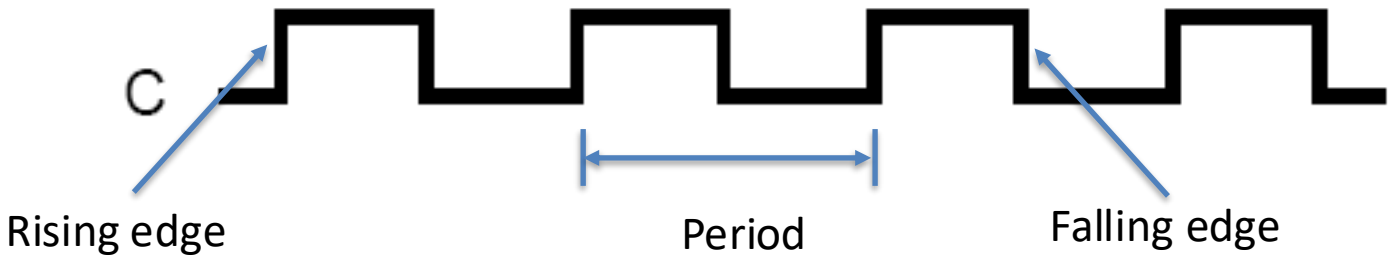

D Latch

# D flip flop: Falling Edge Trigger



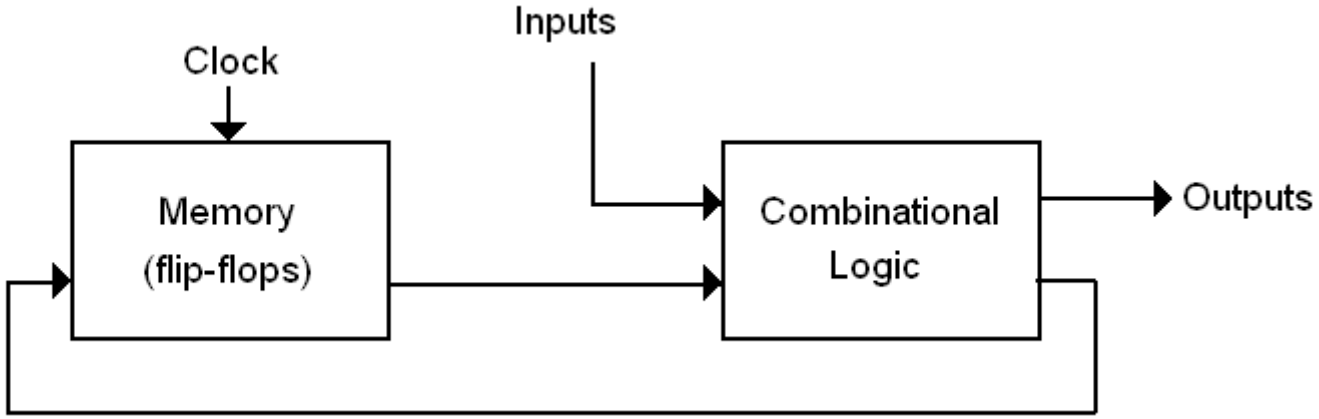



# Edge-triggering




- All changes to state happen at one point in the clock cycle (either rising edge or falling edge).

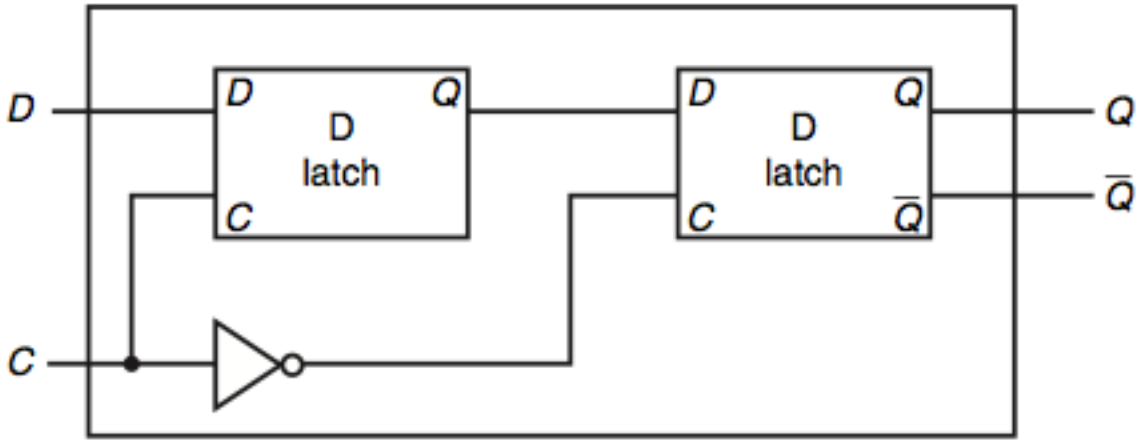
# Memory



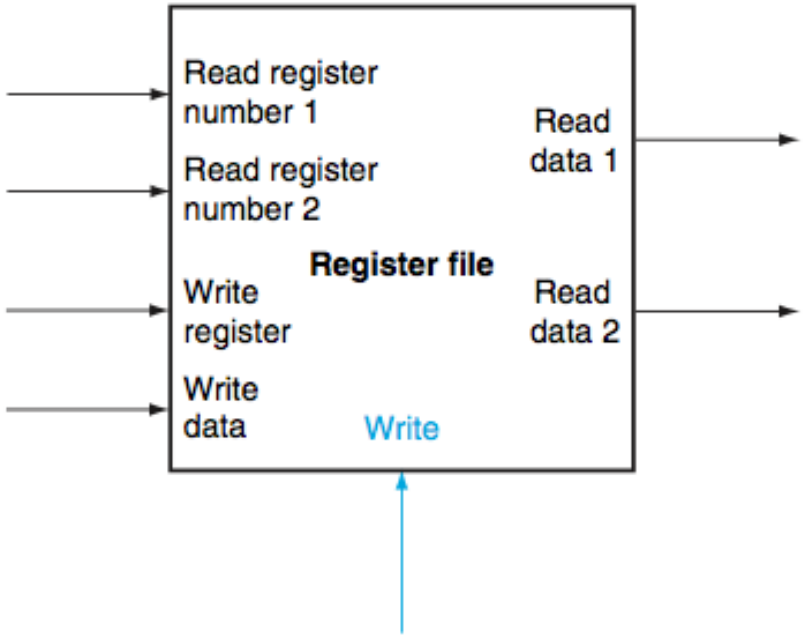

- Can save the results of combinational logic (think the ALU)
- Registers are (multi-bit) flip-flops!

# Registers

- Each 64-bit register will consist of 64 1-bit D flip-flops
- Each D flip-flop holds 1 bit of information
- The same clock signal C is connected each flip-flop (why?)

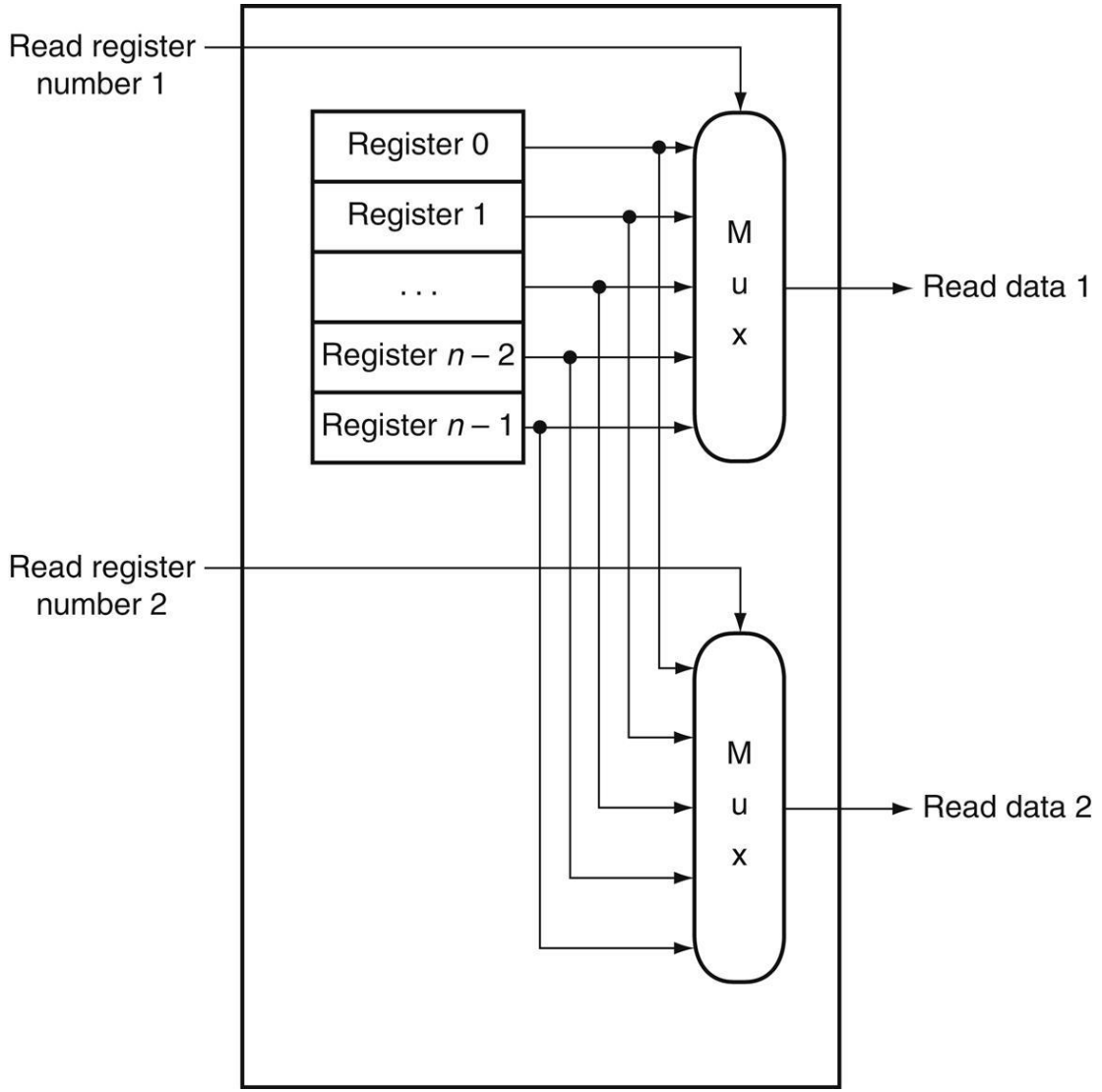


# Register File

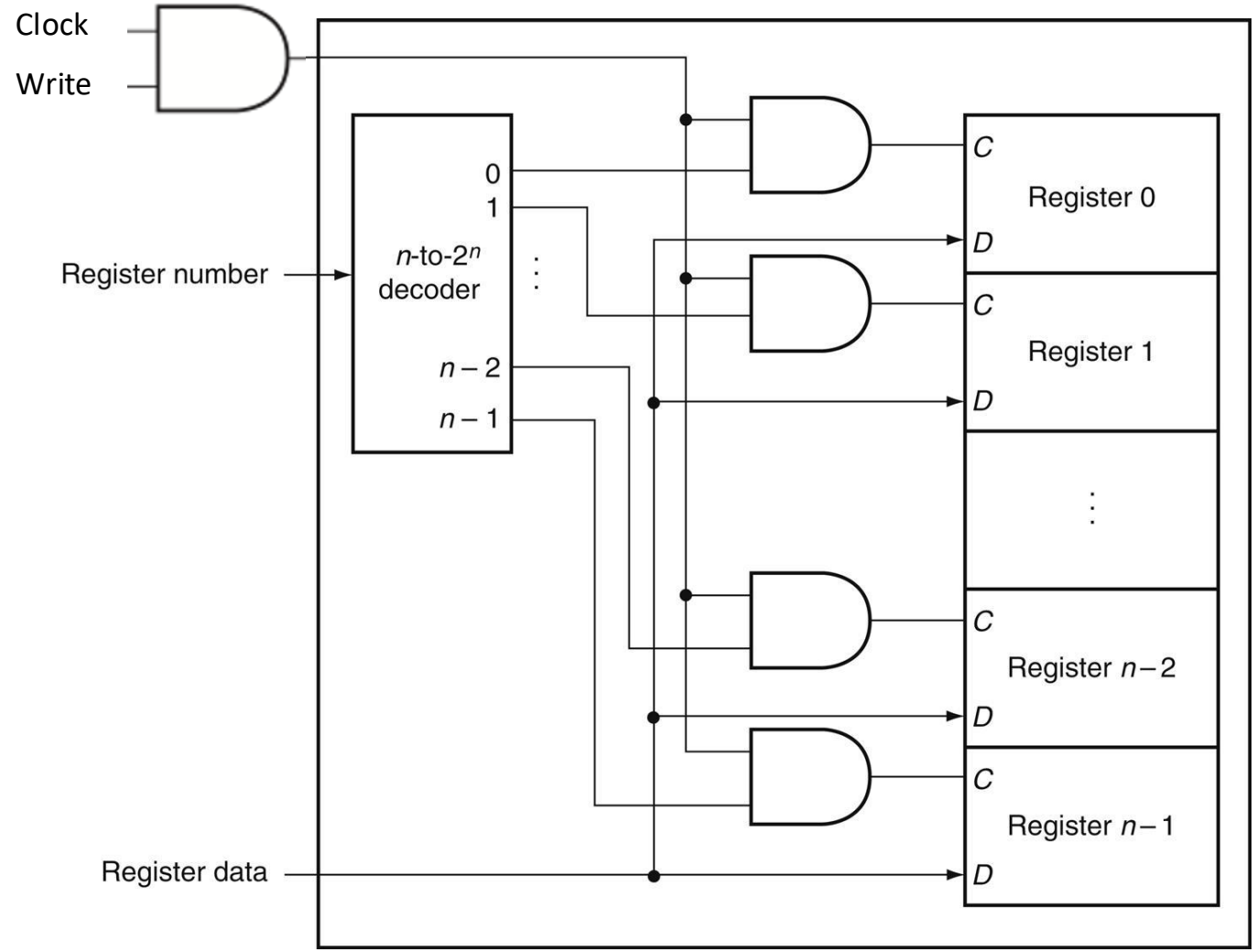



- Set of registers that can be written/read by supplying a register number
- RISC-V has a register file with thirty-two 32-bit registers

# Read two register values



# Write one register value




*The image is not quite correct. It should be a lg n to n decoder

In RISC-V, we have 32 registers so we need a 5-to-32 decoder, not a 32-to-4294967296 decoder!

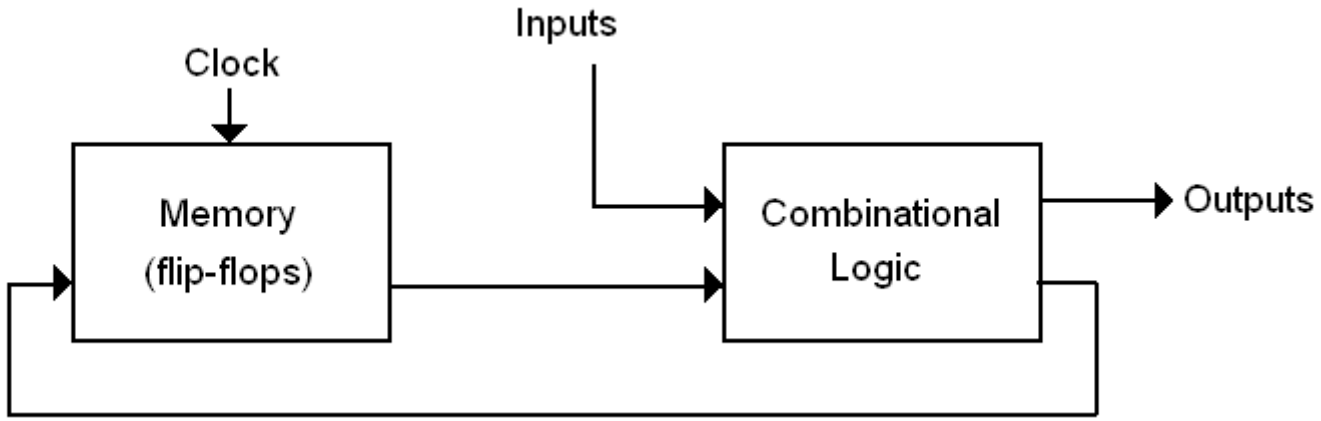
# What will happen if we read and write to a register in the same clock cycle, as in `add t0, t0, s0`

A. The read will get the original value

B. The read will get the just written value

C. It is ambiguous

D. None of the above

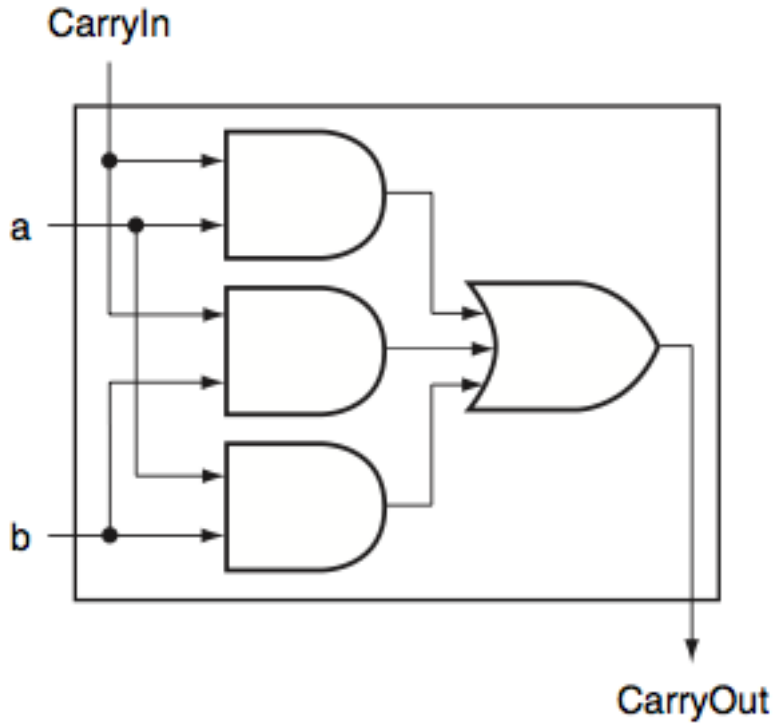


# Register Questions

# Speed of Combinational Circuits

- Assume each gate takes a certain amount of time for the signal to pass through.
- *Gate Delay* is measured by counting the number of gates along a path.
- **Note that all non-sequential gates operate in parallel**

# What is the gate delay to calculate carry-out with this circuit?




A. 1

B. 2

C. 3

D. 4

E. None of the above

# Minimum clock cycle length

- Minimum clock cycle length is determined by
  - The gate delay of the combinational logic
  - The propagation delay of the flip flop (how long does it take for the output to appear after the flip flop's state is changed)
  - The setup time for the flip flop (how long does the value have to be stable before the falling edge of the clock (for falling edge-triggered flip flops)